

**Standard Procedure
for
Calibration

of
Weight class F1,
1 mg to 20 kg**

Issued on January, 2004
Approved by NIMT



1. Calibration Description:

Parameter	Performance Specification	Test Method	Reference Weight
Weights : Conventional mass	- Accuracy class F1 - Range 1 mg – 20 kg	Double Substitutions (ABBA)	Class E2

2. Preliminary Operations for:

- 2.1 Weight (see in appendix A) and OIML R 111-2 requirement checklist (see in form A)
- 2.2 Mass comparator (see in appendix B)

3. Environment conditions :

- 3.1 Temperature: The temperature must be controlled and measured during calibration within ± 1.5 °C per hour with a maximum of ± 20 °C per 12 hours.
- 3.2 Relative humidity: The relative humidity must be controlled and measured during calibration within 40 to 60 % with a maximum of ± 15 % per 4 hours.

4. Calibration Process :

- 4.1 Place the reference weight (r) on the pan of the mass comparator, then wait until the indicated value is stable. Notice the number of seconds it takes. Tare and record the mass comparator reading as *Ir1i*.
- 4.2 Remove the reference weight from the pan of the mass comparator.
- 4.3 Place the test weight (t) on the pan of mass comparator, wait for the same amount of time as 4.1 and then record the mass comparator reading as *It1i*.
- 4.4 Remove the test weight from the pan of the mass comparator.
- 4.5 Place the test weight on the pan of mass comparator, wait for the same amount of time as 4.1 and then record the mass comparator reading as *It2i*.
- 4.6 Remove the test weight from the pan of the mass comparator.
- 4.7 Place the reference weight on the pan of the mass comparator, wait for the same amount of time as 4.1 and then record the mass comparator reading as *Ir2i*, (the first weighing cycle is completed.)

- 4.8 Repeat the step 4.1 through 4.7 without zero setting, until the results of desired weighing cycles are obtained. (normally 3 weighing cycles)

Indication Reading Data

	A	B	B	A
<i>i</i>	<i>Ir1i</i>	<i>It1i</i>	<i>It2i</i>	<i>Ir2i</i>

Where;

Ir1i = The mass comparator indication when the reference weight is first placed on the weighing pan of the *i* cycles.

Ir2i = The mass comparator indication when the reference weight is later placed on the weighing pan of the *i* cycles.

It1i = The mass comparator indication when the test weight is first placed on the weighing pan of the *i* cycles.

It2i = The mass comparator indication when the test weight is later placed on the weighing pan of the *i* cycles.

5. Calculation of Measurement results

- 5.1 The results are processed as follows:

$$\Delta I_i = \frac{(I_{t1i} + I_{t2i} - I_{r1i} - I_{r2i})}{2} \quad (i = 1, 2, \dots, n) \quad (1)$$

$$\Delta \bar{I} = \frac{1}{n} \sum_{i=1}^n \Delta I_i \quad (2)$$

$$\Delta \overline{m_c} = f \times \Delta \bar{I} + m_{cr} C_i \quad (3)$$

Where;

- ΔI_i = The difference of mass comparator indications between the test and reference weight.
- $\Delta \bar{I}$ = The average difference of mass comparator indications between the test and reference weight.
- $\Delta \bar{m}_c$ = The average difference of conventional mass between the test and reference weight.
- f = Scale factor (see in appendix B)
- m_{cr} = Conventional mass value of reference weight.
- C_i = $(\rho_{ai} - \rho_0) \times \left[\frac{1}{\rho_t} - \frac{1}{\rho_r} \right]$
- ρ_{ai} = The air density during calibration (from CIPM formula, see in appendix C).
- ρ_0 = Conventional air density = 1.2 kg.m^{-3}
- ρ_{mt} = Density of the test weight (see in appendix D).
- ρ_{mr} = Density of reference weight (from certificate of manufacturer)
- if $|C_i| \leq \frac{1}{3} \frac{U}{m_0}$ the term $m_{cr}C_i$ can be omitted.
- Where U = Expanded uncertainty.
- m_0 = Nominal value of the weight.

5.2 Model of Evaluation for conventional mass

$$m_{ct} = m_{cr} + \Delta \bar{m}_c + \delta m_b + \delta m_{ba} + \delta m_D \quad (4)$$

Where;

- m_{ct} = Conventional mass value of the test weight to be measured.
- m_{cr} = Conventional mass value of the reference weight.
- $\Delta \bar{m}_c$ = The average difference of conventional mass between the test and reference weight.
- δm_b = The air buoyancy correction during calibration
- δm_{ba} = The effect of the balance during calibration
- δm_D = The effect of drift of the reference weight

6. Uncertainty calculation

1. The standard uncertainty of the reference weight $u(m_{cr})$ (Type B)

$$u(m_{cr}) = \frac{U}{k} \quad (5)$$

U = The uncertainty of the reference weight from certificate
k = The coverage factor

If a combination of the reference weights are used for a mass comparison and their covariances are not known, a correlation coefficient of 1 can be assumed. This will lead to linear summation of uncertainties.

$$U(m_{cr}) = \sum_i u(m_{cri}) \quad (6)$$

Where $u(m_{cri})$ is the standard uncertainty of the reference weight. This is an upper limit for the uncertainty.

2. The standard uncertainty of the weighing process u_w (type A)

$$u_w(\overline{\Delta m_c}) = \frac{S_p}{\sqrt{n}} \quad (7)$$

S_p = Pooled standard deviation of the weighing process (see in appendix B).
n = Number of weighing cycles

3. The standard uncertainty of air buoyancy correction $u_{\delta m_b}$

$$u_{\delta m_b}^2 = \left[m_{cr} \frac{\rho_r - \rho_t}{\rho_r \rho_t} u_{\rho_a} \right]^2 + [m_{cr} (\rho_a - \rho_0)]^2 \frac{u_{\rho_t}^2}{\rho_t^4} - m_{cr}^2 (\rho_a - \rho_0) [(\rho_a - \rho_0) + 2(\rho_{01} - \rho_a)] \frac{u_{\rho_r}^2}{\rho_r^4} \quad (8)$$

Where;

m_{cr} = Conventional mass of the reference weight

ρ_r = Density of the reference weight, kg.m^{-3}

ρ_t = Density of the test weight, kg.m^{-3}

u_{ρ_a} = Standard uncertainty of the air density, kg.m^{-3}

If the air density is not measured and the average air density for the site is used, then the uncertainty for the air density is to be estimated as

$$u(\rho_a) = \frac{0.12}{\sqrt{3}}$$

A lower value of uncertainty may be used if supporting data can be provided.

At sea level the density of air should be assumed to be 1.2 kg/m^3

u_{ρ_r} = Standard uncertainty of the reference weight density, kg.m^{-3}

ρ_a = The air density during the calibration, kg.m^{-3}

ρ_{a1} = The air density during the (previous) calibration
of the reference weight, kg.m^{-3}

ρ_o = Air density at conventional condition ($\rho_o = 1.2$), kg.m^{-3}

ρ_t = The test weight density
(according to a material specification), kg.m^{-3}

u_{ρ_t} = Standard uncertainty of the test weight density, kg.m^{-3}

ρ_r = The reference weight density, kg.m^{-3}

u_{ρ_r} = Standard uncertainty of the reference weight density, kg.m^{-3}

4. The standard uncertainty of the balance $u_{\delta m_{ba}}$ (Type B)

$$u_{\delta m_{ba}} = \sqrt{u_s^2 + u_d^2 + u_E^2 + u_{ma}^2} \quad (9)$$

u_s = Standard uncertainty due to sensitivity of the balance

$$u_s^2 = (\overline{\Delta m_c})^2 \left[\frac{u_{(ms)}^2}{m_s^2} + \frac{u_{(\Delta I_s)}^2}{\Delta I_s^2} \right] \quad (9-1)$$

m_s = The sensitivity weight

u_{ms} = The standard uncertainty of sensitivity mass

ΔI_s = The change in the indication of the mass comparator due to the sensitivity weight

$u_{(\Delta I_s)}$ = The standard uncertainty of ΔI_s

u_d = The standard uncertainty due to the display resolution of the mass comparator

$$u_d = \left(\frac{d/2}{\sqrt{3}} \right) \times \sqrt{2} \quad (9-2)$$

d = Resolution of the mass comparator

u_E = The standard uncertainty due to eccentric loading

$$u_E = \frac{(d1/d2) \times D}{2\sqrt{3}} \quad (9-3-1)$$

or

$$u_E = \frac{|\Delta I_1 - \Delta I_2|}{\sqrt{3}} \quad (\text{For Balance with Automatic mass exchange machine}) \quad (9-3-2)$$

D = The difference between max. and min. values from the eccentricity test

$d1$ = The estimated distance between the centers of the weight.

$d2$ = The distance from the center of the load receptor to one of the corners.

$\Delta I_1 - \Delta I_2$ = The indication difference between two weights may be different when the positions are interchanged

u_{ma} = The standard uncertainty due to magnetism (9-4)
(Assumed to be zero)
Because the masses satisfy the requirement of the recommendation (OIML /CD R 111)

5. The standard uncertainty of drift of reference standard

The uncertainties for long term stability are estimated from the deviation of the present calibrations and the previous calibrations with 2 years and assumed the rectangular distribution

$$u_{\delta m_D} = \frac{\text{Drift}}{2\sqrt{3}} \quad (10)$$

Drift = estimated from the deviation of the present calibrations and the previous calibrations with 2 years

Combined standard uncertainty of the test weight, $u_c(m_t)$

$$u_c(m_t) = \sqrt{u^2(m_r) + u_W^2(\Delta \overline{m}) + u_{\delta m_b}^2 + u_{\delta m_{ba}}^2 + u_{\delta m_D}^2} \quad (11)$$

The expanded uncertainty: $U(m_t)$

$$U(m_t) = k \cdot u_c(m_t) \quad (12)$$

$k = 2$

Note If a pooled standard deviation of the weighing process is not know, the number of measurements can not be increased up to 10 and the uncertainty due to weighing process (u_w) is the dominant in the uncertainty analysis, $u_w(\Delta \overline{m}) > u_c(m_t) / 2$ then the coverage factor k should be calculated from the t distribution assuming 95.5% confidence level and effective degrees of freedom (ν_{eff})

$$\nu_{eff} = (n-1) \times \frac{u_c^4(m_t)}{u_w^4(\Delta \overline{m})}$$

ν_{eff}	1	2	3	4	5	6	8	10	20	∞
k	13.97	4.53	3.31	2.87	2.65	2.52	2.37	2.28	2.13	2.00

According to OIML/CDR-111, the expanded uncertainty of the conventional mass for each weight shall be less than or equal to one third of the MPE. Normally the calculating expanded uncertainties (equation 7) are less than one third of the MPE. The reporting expanded uncertainty of the Calibration Laboratory shall therefore be one third of the MPE.

The uncertainty budget of the test weight

Quantity X_i	Estimate (x_i) (g)	Standard uncertainty $u(x_i)$ (mg)	Probability distribution	Sensitivity Coefficient (C_i)	Uncertainty Contribution (mg)
m_r	_____	(5), (6)	_____	_____	_____
$\Delta \bar{m}_c$	_____	(7)	_____	_____	_____
δm_b	_____	(8)	_____	_____	_____
δm_{ba}	_____	(9)	_____	_____	_____
δm_D	_____	(10)	_____	_____	_____
m_t	_____ kg + _____ mg		normal	Combined standard uncertainty	(11)
				Expanded uncertainty	(12)
				Reported uncertainty	_____

FORM A

Check conditions of weight before calibration

Description Nominal weight		Smooth and without scratch	Surface does not appear to be porous	Remark for Fail			
				Top surface	Bottom surface	surface	
	P						
	F						
	P						
	F						
	P						
	F						
	P						
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APPENDIX A**Preliminary Operations for weight****Procedure:**

1. The weight must not be grasped in the hand. It is essential to use tweezers, a cloth or special gloves.
2. The weight must be handled and stored in such a way that they stay clean. Before calibration dust and any foreign particles should be removed using, for example, air puffer (clean gas, not compressed air) or a soft brush. Care should be taken to ensure that the surface properties of the weight are not changed, for example, scratching.
3. The weight must be kept in the room for thermal stabilization, the mandatory minimum times required for temperature equalization are shown in Table 1.

Table 1: Thermal stabilization

ΔT^*	Weight size	Class E2	Class F1	Class F2
$\pm 20^\circ\text{C}$	10, 20 kg	27 hrs	12 hrs	3 hrs
	1, 2, 5 kg	12 hrs	6 hrs	2 hrs
	100, 200, 500 g	5 hrs	3 hrs	1 hr
	10, 20, 50 g	2 hrs	1 hr	1 hr
	<10 g	1 hr	1 hr	0.5 hr
$\pm 5^\circ\text{C}$	10, 20 kg	18 hrs	4 hrs	1 hr
	1, 2, 5 kg	8 hrs	3 hrs	1 hr
	100, 200, 500 g	4 hrs	2 hrs	0.5 hr
	10, 20, 50 g	1 hr	1 hr	0.5 hr
	<10 g	0.5 hr	0.5 hr	0.5 hr
$\pm 2^\circ\text{C}$	10, 20 kg	10 hrs	1 hr	0.5 hr
	1, 2, 5 kg	5 hrs	1 hr	0.5 hr
	100, 200, 500 g	3 hrs	1 hr	0.5 hr
	<100 g	1 hr	1 hr	0.5 hr
$\pm 0.5^\circ\text{C}$	10, 20 kg	1 hr	0.5 hr	0.5 hr
	1, 2, 5 kg	1 hr	0.5 hr	0.5 hr
	100, 200, 500 g	1 hr	0.5 hr	0.5 hr
	<100 g	0.5 hr	0.5 hr	0.5 hr

ΔT^* = Initial difference between weight temperature and ambient temperature

4. If the weight contains significant amounts of dirt, which can not be removed by the aforementioned methods, the weight or some parts of it may be cleaned with alcohol, distilled water or other solvents. Weights with internal cavities should normally not be immersed in the solvent to avoid the possibility that the fluid will penetrate the opening. If there is a need to monitor the stability of a weight in use, the mass of the weight should, if possible, be determined before cleaning

After cleaning, the weight will need time to stabilize. Typical stabilization times are shown in Table 2:

Table 2: Stabilization time after cleaning

Weight Class	E2	F1	F2 to M3
After cleaning with alcohol	3-6 days	1-2 days	1 hour
After cleaning with distilled water	2-3 days	1 days	1 hour

APPENDIX B**For Mass comparator****1. Preliminary**

- 1.1 The mass comparator must be switched on before use some period of time depends on the manual operation document to ensure that ready to use.
- 1.2 The level of the mass comparator must be checked and adjusted correctly.
- 1.3 Before beginning calibration process, a weight must be placed and removed on the pan of the mass comparator several times to ensure that the least significant digit of the mass comparator indication is stable (the weight should closed to the maximum capacity).
- 1.4 To check the standard deviation during calibration procedure, taking into account the limit control of each mass comparator. If the standard deviation is larger than the limit control, the calibration must be cancelled in order to solve the problem.

2. Pooled standard deviation

The standard uncertainty of the weighing process can be determined from a pooled standard deviation, which obtained from earlier ABBA measurements model under similar conditions. The reference weights class E2 are chosen as a reference (reading as A), the OIML weights class F1 are selected as a test weight (reading as B). The nominal value of the weights must be within the range of the mass comparator. The pooled standard deviation should be performed at maximum capacity of the mass comparator, try another lower capacities and choose the worse one. The pooled standard deviation procedure is as follow:

- 2.1 Place the reference weight on the pan of the mass comparator, then wait until the indicated value is stable and notice the number of seconds it takes. Tare the mass comparator and record value reading as A1 ;
- 2.2 Remove the reference weight from the pan of the mass comparator.
- 2.3 Place the test weight on the pan of mass comparator, wait for the same period of time as 2.1 and then record the mass comparator indication, reading as B1 ;
- 2.4 Remove the test weight from the pan of the mass comparator wait for same period of time as 2.1
- 2.5 Place the test weight on the pan of mass comparator, wait for the same period of time as 2.1 and then record the mass comparator indication, reading as B2 ;

- 2.6 Remove the test weight from the pan of the mass comparator wait for same period of time as 2.1
- 2.7 Place the reference weight on the pan of mass comparator, wait for the same period of time as 2.1 and then record the mass comparator indication, reading as $A2_i$ (The first weighing cycle is completed)
- 2.8 Repeat the step 2.1 through 2.7 without zero setting until the results of ten weighing cycles are obtained.

Indication reading data

i	A1 _i	B1 _i	B2 _i	A2 _i	$\Delta I_p = (B1_i + B2_i)2 - (A1_i + A2_i)/2$
1	A1 ₁	B1 ₁	B2 ₁	A2 ₁	ΔI_{p1}
2	A1 ₂	B1 ₂	B2 ₂	A2 ₂	ΔI_{p2}
3	A1 ₃	B1 ₃	B2 ₃	A2 ₃	ΔI_{p3}
4	A1 ₄	B1 ₄	B2 ₄	A2 ₄	ΔI_{p4}
5	A1 ₅	B1 ₅	B2 ₅	A2 ₅	ΔI_{p5}
6	A1 ₆	B1 ₆	B2 ₆	A2 ₆	ΔI_{p6}
7	A1 ₇	B1 ₇	B2 ₇	A2 ₇	ΔI_{p7}
8	A1 ₈	B1 ₈	B2 ₈	A2 ₈	ΔI_{p8}
9	A1 ₉	B1 ₉	B2 ₉	A2 ₉	ΔI_{p9}
10	A1 ₁₀	B1 ₁₀	B2 ₁₀	A2 ₁₀	ΔI_{p10}
					$\overline{\Delta I_p}$ = average of ΔI_{p_i}

Calculation

The standard deviation (S)

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\Delta I_{p_i} - \overline{\Delta I_p})^2}$$

3. Scale factor determination

The scale factor of the mass comparator should be calibrated with a sensitivity weight (m_s), the procedure due to scale factor determination is as follow;

- 3.1 Place the tare weight, the nominal value close to the maximum capacity of the mass comparator, on the pan of the mass comparator wait until the indication is stable then press tare and record the indication value as $S1_i$.
- 3.2 Place the sensitivity weight (m_s), normally the nominal value of the sensitivity weight should be closed to the $2/3$ of the maximum permissible error of the test weight class F1, on the pan of mass comparator and wait until the indication is stable then record the indication value as $T1_i$.
- 3.3 Remove the sensitivity weight from the pan of the mass comparator and wait for the same period of time as in 3.2
- 3.4 Place the sensitivity weight (m_s) on the pan of the mass comparator and wait the same period of time as 3.2. Record the indication value as $T2_i$.
- 3.5 Remove the sensitivity weight (m_s) on the pan of the mass comparator and wait the same period of time as 3.2. Record the indication value as $S2_i$ (The first weighing cycle is completed)
- 3.6 Repeat the step 3.1 through 3.5 until the results of ten weighing cycles are obtained.

Indication reading data

i	$S1_i$	$T1_i$	$T2_i$	$S2_i$	$\Delta Is_i = (T1_i + T2_i)2 - (S1_i + S2_i)/2$
1	$S1_1$	$B1_1$	$B2_1$	$S2_1$	ΔIs_1
2	$S1_2$	$B1_2$	$B2_2$	$S2_2$	ΔIs_2
3	$S1_3$	$B1_3$	$B2_3$	$S2_3$	ΔIs_3
4	$S1_4$	$B1_4$	$B2_4$	$S2_4$	ΔIs_4
5	$S1_5$	$B1_5$	$B2_5$	$S2_5$	ΔIs_5
6	$S1_6$	$B1_6$	$B2_6$	$S2_6$	ΔIs_6
7	$S1_7$	$B1_7$	$B2_7$	$S2_7$	ΔIs_7
8	$S1_8$	$B1_8$	$B2_8$	$S2_8$	ΔIs_8
9	$S1_9$	$B1_9$	$B2_9$	$S2_9$	ΔIs_9
10	$S1_{10}$	$B1_{10}$	$B2_{10}$	$S2_{10}$	ΔIs_{10}
					$\overline{\Delta Is} = \text{average of } \Delta Is_i$

Calculation

Calculate the scale factor (f)

$$f = \frac{\overline{\Delta I_s}}{m_s}$$

Where,

$\overline{\Delta I_s}$ = The average of the indication of the mass comparator due to sensitivity weight

m_s = The mass of the sensitivity weight

APPENDIX C**Approximation formula for air density**

The most accurate formula of air density is the CIPM formula

An approximate formula may also be used:

$$\rho_a = \frac{0.34848p - 0.009024(rh) \times \exp(0.0612t)}{273.15 + t}$$

where the density of air, ρ_a is obtained in kg.m^{-3} , when the pressure, p is given in mbar or hPa, the relative humidity, rh expressed as a fraction, and the temperature, t in $^{\circ}\text{C}$

APPENDIX D**Estimating the density of weight**

The Estimating the density based on the known composition of the alloy from which the weight is manufactured is one of the Methods for determining density. Obtain the composition of the alloy from the supplier of the weight in question. Find the density value from a physics/chemistry handbook that has tables of density as a function of the concentration of alloying elements. Use the handbook density value and apply the uncertainty value from Table 3 below. For class E2 to M2 weights the “assumed density” values in Table 3 below are adequate. The density of M3 weights is usually of no concern.

Table 3: List of alloys most commonly used for weights

Alloy / material	Assumed density	Uncertainty ($k=2$)
Platinum	21400 kg/m ³	±150 kg/m ³
Nickel silver	8600 kg/m ³	±170 kg/m ³
Brass	8400 kg/m ³	±170 kg/m ³
Stainless steel	7950 kg/m ³	±140 kg/m ³
Carbon steel	7700 kg/m ³	±200 kg/m ³
Iron	7800 kg/m ³	±200 kg/m ³
Cast iron	7700 kg/m ³	±400 kg/m ³
Cast iron	7100 kg/m ³	±600 kg/m ³
Aluminum	2700 kg/m ³	±130 kg/m ³